

# Probability

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## Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
```

```
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

---

Most of the content in this chapter should be review from UC Davis Epi 202.

# 1 Core properties of probabilities

## 1.1 Defining probabilities

**Definition 1.1** (Probability measure). A **probability measure**, often denoted  $\Pr()$  or  $P()$ , is a function whose domain is a  $\sigma$ -algebra<sup>1</sup> of possible outcomes,  $\mathcal{S}$ , and which satisfies the following properties:

1. For any statistical event  $A \in \mathcal{S}$ ,  $\Pr(A) \geq 0$ .
2. The probability of the union of all outcomes ( $\Omega \stackrel{\text{def}}{=} \cup \mathcal{S}$ ) is 1:

$$\Pr(\Omega) = 1$$

3. The probability of the union of disjoint events,  $A_1 \cup A_2 : A_1 \cap A_2 = \emptyset$ , is equal to the sum of their probabilities:

$$\Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2)$$

---

**Theorem 1.1.** If  $A$  and  $B$  are statistical events and  $A \subseteq B$ , then  $\Pr(A \cap B) = \Pr(A)$ .

---

<sup>1</sup><https://en.wikipedia.org/wiki/%CE%A3-algebra>

---

*Proof.* Left to the reader for now. □

---

**Theorem 1.2.**

$$\Pr(A) + \Pr(\neg A) = 1$$

---

*Proof.* By properties 2 and 3 of Definition 1.1. □

---

**Corollary 1.1.**

$$\Pr(\neg A) = 1 - \Pr(A)$$

---

*Proof.* By Theorem 1.2 and algebra. □

---

**Corollary 1.2.** *If the probability of an outcome  $A$  is  $\Pr(A) = \pi$ , then the probability that  $A$  does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

---

*Proof.* Using Corollary 1.1:

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) \\ &= 1 - \pi\end{aligned}$$

---

□

## 2 Random variables

### 2.1 Binary variables

**Definition 2.1** (binary variable). A **binary variable** is a random variable which has only two possible values in its range.

**Exercise 2.1** (Examples of binary variables). What are some examples of binary variables in the health sciences?

---

*Solution.* Examples of binary outcomes include:

- exposure (exposed vs unexposed)
  - disease (diseased vs healthy)
  - recovery (recovered vs unrecovered)
  - relapse (relapse vs remission)
  - return to hospital (returned vs not)
  - vital status (dead vs alive)
-

## 2.2 Count variables

**Definition 2.2** (Count variable). A **count variable** is a random variable whose possible values are some subset of the non-negative integers; that is, a random variable  $X$  such that:

$$\mathcal{R}(X) \in \mathbb{N}$$

---

**Exercise 2.2.** What are some examples of count variables?

---

*Solution.*

- Number of fish in a pond
  - Number of cyclones per season
  - Seconds of tooth-brushing per session (if rounded)<sup>2</sup>
  - Infections per person-year
  - Visits to ER per person-month
  - Car accidents per 1000 miles driven
- 

**Definition 2.3** (Exposure magnitude). For many count outcomes, there is some sense of an **exposure magnitude**, such as **population size**, or **duration of observation**, which multiplicatively rescales the expected (mean) count.

---

**Exercise 2.3.** What are some examples of exposure magnitudes?

---

*Solution.*

Table 1: Examples of exposure units

| outcome            | exposure units                              |
|--------------------|---------------------------------------------|
| disease incidence  | number of individuals exposed; time at risk |
| car accidents      | miles driven                                |
| worksite accidents | person-hours worked                         |
| population size    | size of habitat                             |

---

Exposure units are similar to the number of trials in a binomial distribution, but **in non-binomial count outcomes, there can be more than one event per unit of exposure.**

We can use  $t$  to represent continuous-valued exposures/observation durations, and  $n$  to represent discrete-valued exposures.

---

**Definition 2.4** (Event rate).

For a count outcome  $Y$  with exposure magnitude  $t$ , the **event rate** (denoted  $\lambda$ ) is defined as the mean of  $Y$  divided by the exposure magnitude. That is:

$$\mu \stackrel{\text{def}}{=} E[Y|T = t]$$

$$\lambda \stackrel{\text{def}}{=} \frac{\mu}{t} \tag{1}$$

---

<sup>2</sup><https://pubmed.ncbi.nlm.nih.gov/35587489/>

Event rate is somewhat analogous to odds in binary outcome models; it typically serves as an intermediate transformation between the mean of the outcome and the linear component of the model. However, in contrast with the odds function, the transformation  $\lambda = \mu/t$  is *not* considered part of the Poisson model's link function, and it treats the exposure magnitude covariate differently from the other covariates.

---

**Theorem 2.1** (Transformation function from event rate to mean). *For a count variable with mean  $\mu$ , event rate  $\lambda$ , and exposure magnitude  $t$ :*

$$\therefore \mu = \lambda \cdot t \tag{2}$$


---

*Solution.* Start from definition of event rate and use algebra to solve for  $\mu$ .

---

Equation 2 is analogous to the inverse-odds function for binary variables.

---

**Theorem 2.2.** *When the exposure magnitude is 0, there is no opportunity for events to occur:*

$$E[Y|T = 0] = 0$$


---

*Proof.*

$$E[Y|T = 0] = \lambda \cdot 0 = 0$$

□

---

#### Probability distributions for count outcomes

- [Poisson distribution](#)
  - [Negative binomial distribution](#)
- 

### 3 Key probability distributions

---

Some distributions are typically used for outcome models (Table 2); other distributions are typically used for test statistics (Table 3).

Table 2: Distributions typically used for outcome models

| Distribution | Uses                                        |
|--------------|---------------------------------------------|
| Bernoulli    | Binary outcomes                             |
| Binomial     | Sums of Bernoulli outcomes                  |
| Poisson      | unbounded count outcomes                    |
| Geometric    | Counts of non-events before an event occurs |

| Distribution      | Uses                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------|
| Negative binomial | Mixtures of Poisson distributions, counts of non-events until a given number of events occurs |
| Normal (Gaussian) | Continuous outcomes without a more specific distribution                                      |
| exponential       | Time to event outcomes                                                                        |
| Gamma             | Time to event outcomes                                                                        |
| Weibull           | Time to event outcomes                                                                        |
| Log-normal        | Time to event outcomes                                                                        |

Table 3: Distributions typically used for test statistics

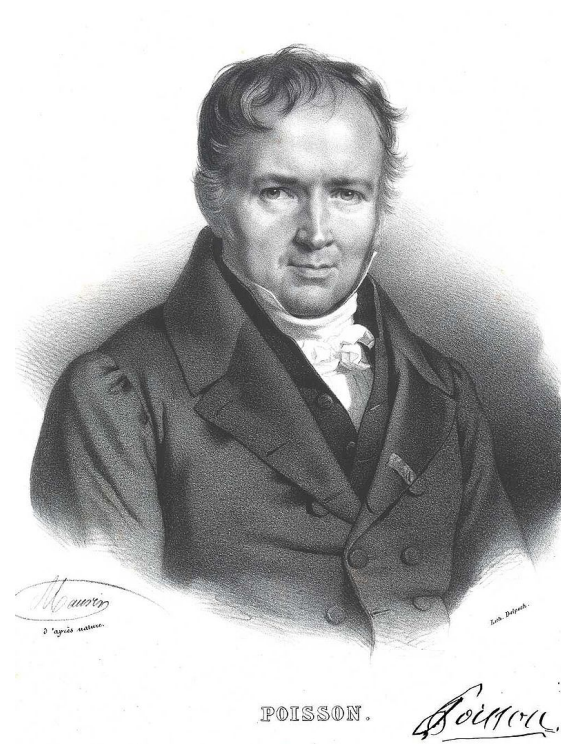
| Distribution          | Uses                                                                                             |
|-----------------------|--------------------------------------------------------------------------------------------------|
| $\chi^2$              | Regression comparisons (asymptotic), contingency table independence tests, goodness-of-fit tests |
| $F$                   | Gaussian model comparisons (exact)                                                               |
| $Z$ (standard normal) | Proportions, means, regression coefficients (asymptotic)                                         |
| $T$                   | Means, regression coefficients in Gaussian outcome models (exact)                                |

### 3.1 The Bernoulli distribution

**Definition 3.1** (Bernoulli distribution). The **Bernoulli distribution** family for a random variable  $X$  is defined as:

$$\begin{aligned}
 \Pr(X = x) &= 1_{x \in \{0,1\}} \pi^x (1 - \pi)^{1-x} \\
 &= \begin{cases} \pi, & x = 1 \\ 1 - \pi, & x = 0 \end{cases}
 \end{aligned}$$

### 3.2 The Poisson distribution



(a) Siméon Denis Poisson



(b) Les Poissons<sup>a</sup>

<sup>a</sup><https://youtu.be/UoJxBEQRLd0?t=12>

Figure 1: “Les Poissons”

---

**Exercise 3.1.** Define the Poisson distribution.

---

*Solution 3.1.*

*Definition 3.2* (Poisson distribution).

$$P(Y = y) = \frac{\mu^y e^{-\mu}}{y!}, y \in \mathbb{N} \quad (3)$$

(see Figure 2)

---

**Exercise 3.2.** What is the range of possible values for a Poisson distribution?

---

*Solution 3.2.*

$$\mathcal{R}(Y) = \{0, 1, 2, \dots\} = \mathbb{N}$$

---

**Theorem 3.1** (CDF of Poisson distribution).

$$P(Y \leq y) = e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} \quad (4)$$

(see Figure 3)

```

library(dplyr)
pois_dists <- tibble(
  mu = c(0.5, 1, 2, 5, 10, 20)
) |>
  reframe(
    .by = mu,
    x = 0:30
  ) |>
  mutate(
    `P(X = x)` = dpois(x, lambda = mu),
    `P(X <= x)` = ppois(x, lambda = mu),
    mu = factor(mu)
  )

library(ggplot2)
library(latex2exp)

plot0 <- pois_dists |>
  ggplot(
    aes(
      x = x,
      y = `P(X = x)`,
      fill = mu,
      col = mu
    )
  ) +
  theme(legend.position = "bottom") +
  labs(
    fill = latex2exp::TeX("$\\mu$"),
    col = latex2exp::TeX("$\\mu$"),
    y = latex2exp::TeX("$\\Pr_{\\mu}(X = x)$")
  )

plot1 <- plot0 +
  geom_segment(yend = 0) +
  facet_wrap(~mu)

print(plot1)

```

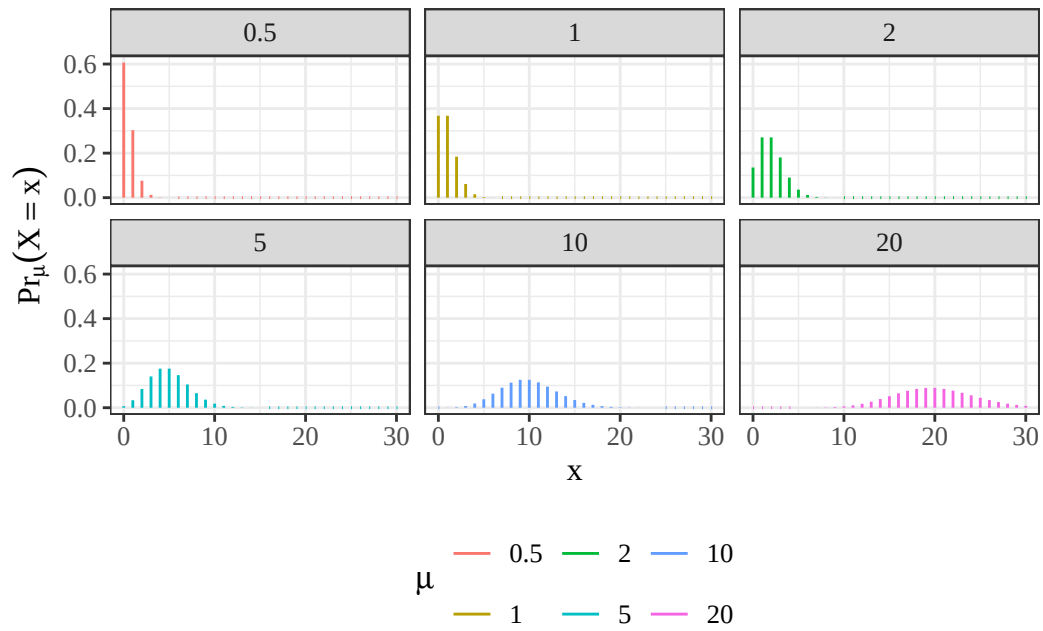


Figure 2: Poisson PMFs, by mean parameter  $\mu$

---

```
library(ggplot2)

plot2 <-
  plot0 +
  geom_step(alpha = 0.75) +
  aes(y = `P(X <= x)`) +
  labs(y = latex2exp::TeX("$\\Pr_{\\mu}(X \\leq x)$"))

print(plot2)
```

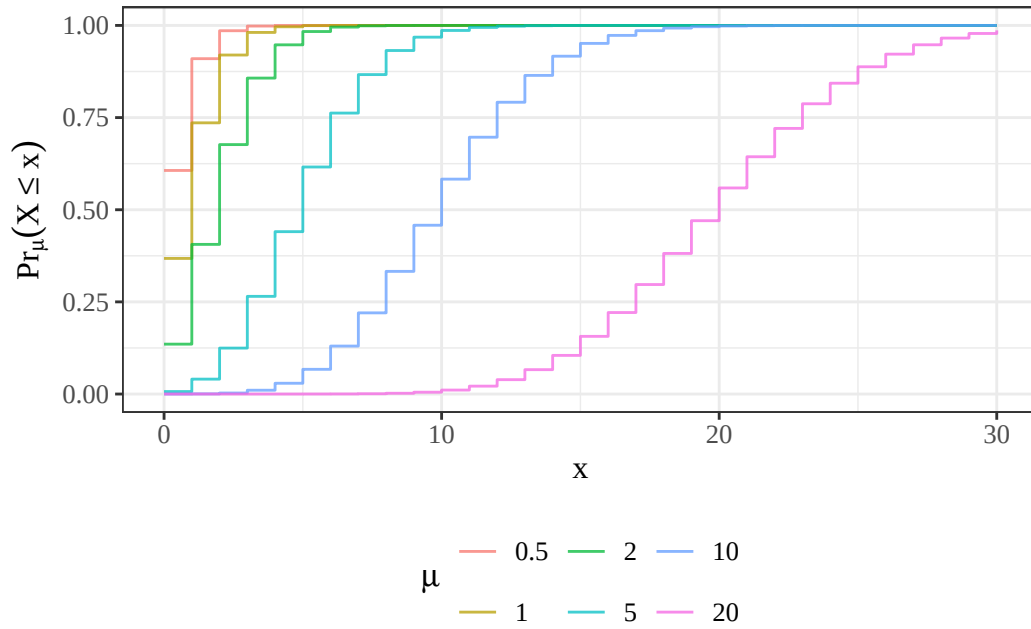


Figure 3: Poisson CDFs

---

**Exercise 3.3** (Poisson distribution functions). Let  $X \sim \text{Pois}(\mu = 3.75)$ .

Compute:

- $P(X = 4 | \mu = 3.75)$
- $P(X \leq 7 | \mu = 3.75)$
- $P(X > 5 | \mu = 3.75)$

---

*Solution.*

- $P(X = 4) = 0.19378$
- $P(X \leq 7) = 0.962379$
- $P(X > 5) = 0.177117$

---

**Theorem 3.2** (Properties of the Poisson distribution). If  $X \sim \text{Pois}(\mu)$ , then:

- $E[X] = \mu$
- $\text{Var}(X) = \mu$
- $P(X = x) = \frac{\mu}{x} P(X = x - 1)$
- For  $x < \mu$ ,  $P(X = x) > P(X = x - 1)$
- For  $x = \mu$ ,  $P(X = x) = P(X = x - 1)$
- For  $x > \mu$ ,  $P(X = x) < P(X = x - 1)$
- $\arg \max_x P(X = x) = \lfloor \mu \rfloor$

**Exercise 3.4.** Prove Theorem 3.2.

---

*Solution.*

$$\begin{aligned}
E[X] &= \sum_{x=0}^{\infty} x \cdot P(X = x) \\
&= 0 \cdot P(X = 0) + \sum_{x=1}^{\infty} x \cdot P(X = x) \\
&= 0 + \sum_{x=1}^{\infty} x \cdot P(X = x) \\
&= \sum_{x=1}^{\infty} x \cdot P(X = x) \\
&= \sum_{x=1}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x!} \\
&= \sum_{x=1}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x \cdot (x-1)!} && \text{[definition of factorial ("!") function]} \\
&= \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!} \\
&= \sum_{x=1}^{\infty} \frac{(\lambda \cdot \lambda^{x-1}) e^{-\lambda}}{(x-1)!} \\
&= \lambda \cdot \sum_{x=1}^{\infty} \frac{(\lambda^{x-1}) e^{-\lambda}}{(x-1)!} \\
&= \lambda \cdot \sum_{y=0}^{\infty} \frac{(\lambda^y) e^{-\lambda}}{(y)!} && \text{[substituting } y \stackrel{\text{def}}{=} x - 1] \\
&= \lambda \cdot 1 && \text{[because PDFs sum to 1]} \\
&= \lambda
\end{aligned}$$

See also <https://statproofbook.github.io/P/poiss-mean>.

For the variance, see <https://statproofbook.github.io/P/poiss-var>.

### Accounting for exposure

If the exposures/observation durations, denoted  $T = t$  or  $N = n$ , vary between observations, we model:

$$\mu = \lambda \cdot t$$

$\lambda$  is interpreted as the “expected event rate per unit of exposure”; that is,

$$\lambda = \frac{E[Y|T = t]}{t}$$

#### ! Important

The exposure magnitude,  $T$ , is *similar* to a covariate in linear or logistic regression. However, there is an important difference: in count regression, **there is no intercept corresponding to  $E[Y|T = 0]$** . In other words, this model assumes that if there is no exposure, there can’t be any events.

**Theorem 3.3.** *If  $\mu = \lambda \cdot t$ , then:*

$$\log \mu = \log \lambda + \log t$$

**Definition 3.3** (Offset). When the linear component of a model involves a term without an unknown coefficient, that term is called an **offset**.

---

**Theorem 3.4.** If  $X$  and  $Y$  are independent Poisson random variables with means  $\mu_X$  and  $\mu_Y$ , their sum,  $Z = X + Y$ , is also a Poisson random variable, with mean  $\mu_Z = \mu_X + \mu_Y$ .

---

*Proof.* See [https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12\\_independent\\_rvs.pdf](https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12_independent_rvs.pdf), Example 3. □

---

### 3.3 The Negative-Binomial distribution

**Definition 3.4** (Negative binomial distribution).

$$P(Y = y) = \frac{\mu^y}{y!} \cdot \frac{\Gamma(\rho + y)}{\Gamma(\rho) \cdot (\rho + \mu)^y} \cdot \left(1 + \frac{\mu}{\rho}\right)^{-\rho}$$

where  $\rho$  is an overdispersion parameter and  $\Gamma(x) = (x - 1)!$  for integers  $x$ .

You don't need to memorize or understand this expression.

As  $\rho \rightarrow \infty$ , the second term converges to 1 and the third term converges to  $\exp\{-\mu\}$ , which brings us back to the Poisson distribution.

---

**Theorem 3.5.** If  $Y \sim \text{NegBin}(\mu, \rho)$ , then:

- $E[Y] = \mu$
- $\text{Var}(Y) = \mu + \frac{\mu^2}{\rho} > \mu$

### 3.4 Weibull Distribution

$$p(t) = \alpha \lambda x^{\alpha-1} e^{-\lambda x^\alpha}$$

$$\lambda(t) = \alpha \lambda x^{\alpha-1}$$

$$S(t) = e^{-\lambda x^\alpha}$$

$$E(T) = \Gamma(1 + 1/\alpha) \cdot \lambda^{-1/\alpha}$$

When  $\alpha = 1$  this is the exponential. When  $\alpha > 1$  the hazard is increasing and when  $\alpha < 1$  the hazard is decreasing. This provides more flexibility than the exponential.

We will see more of this distribution later.

## 4 Characteristics of probability distributions

### 4.1 Probability density function

**Definition 4.1** (probability density). If  $X$  is a continuous random variable, then the **probability density** of  $X$  at value  $x$ , denoted  $f(x)$ ,  $f_X(x)$ ,  $p(x)$ ,  $p_X(x)$ , or  $p(X = x)$ , is defined as the limit of the probability (mass) that  $X$  is in an interval around  $x$ , divided by the width of that interval, as that width reduces to 0.

$$f(x) \stackrel{\text{def}}{=} \lim_{\Delta \rightarrow 0} \frac{P(X \in [x, x + \Delta])}{\Delta}$$

See also Rothman et al. (2021) (Chapter 22, p. 535) and [https://en.wikipedia.org/wiki/Probability\\_density\\_function#Formal\\_definition](https://en.wikipedia.org/wiki/Probability_density_function#Formal_definition)

---

**Theorem 4.1** (Density function is derivative of CDF). *The density function  $f(t)$  or  $p(T = t)$  for a random variable  $T$  at value  $t$  is equal to the derivative of the cumulative probability function  $F(t) \stackrel{\text{def}}{=} P(T \leq t)$ ; that is:*

$$f(t) \stackrel{\text{def}}{=} \frac{\partial}{\partial t} F(t)$$


---

**Theorem 4.2** (Density functions integrate to 1). *For any density function  $f(x)$ ,*

$$\int_{x \in \mathcal{R}(X)} f(x) dx = 1$$


---

## 4.2 Hazard function

**Definition 4.2** (Hazard function, hazard rate, hazard rate function).

The **hazard function**, **hazard rate**, **hazard rate function**, for a random variable  $T$  at value  $t$ , typically denoted as  $h(t)$  <sup>3</sup> or  $\lambda(t)$ , <sup>4</sup> is the conditional density<sup>5</sup> of  $T$  at  $t$ , given  $T \geq t$ . That is:

$$\lambda(t) \stackrel{\text{def}}{=} p(T = t | T \geq t)$$

If  $T$  represents the time at which an event occurs, then  $\lambda(t)$  is the probability that the event occurs at time  $t$ , given that it has not occurred prior to time  $t$ .

---

Table 4: Probability distribution functions

| Name                                   | Symbols            | Definition                         |
|----------------------------------------|--------------------|------------------------------------|
| Probability density function (PDF)     | $f(t), p(t)$       | $p(T = t)$                         |
| Cumulative distribution function (CDF) | $F(t), P(t)$       | $P(T \leq t)$                      |
| Survival function                      | $S(t), \bar{F}(t)$ | $P(T > t)$                         |
| Hazard function                        | $\lambda(t), h(t)$ | $p(T = t   T \geq t)$              |
| Cumulative hazard function             | $\Lambda(t), H(t)$ | $\int_{u=-\infty}^t \lambda(u) du$ |
| Log-hazard function                    | $\eta(t)$          | $\log\{\lambda(t)\}$               |

---

$$f(t) \xleftarrow{\frac{-S'(t)}{S(t)\lambda(t)}} S(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \xleftarrow{\int_{u=0}^t \lambda(u) du} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$f(t) \xrightarrow{\frac{f(t)/\lambda(t)}{\int_{u=t}^{\infty} f(u) du}} S(t) \xrightarrow{-\log S(t)} \Lambda(t) \xrightarrow{\Lambda'(t)} \lambda(t) \xrightarrow{\log\{\lambda(t)\}} \eta(t)$$


---

<sup>3</sup>for example in Dobson and Barnett (2018), Vittinghoff et al. (2012), Klein and Moeschberger (2003), and Kleinbaum and Klein (2012)

<sup>4</sup>for example, in Rothman et al. (2021) and Kalbfleisch and Prentice (2011)

<sup>5</sup>[probability.qmd#def-pdf](#)

### 4.3 Expectation

**Definition 4.3** (Expectation, expected value, population mean ). The **expectation**, **expected value**, or **population mean** of a *continuous* random variable  $X$ , denoted  $E[X]$ ,  $\mu(X)$ , or  $\mu_X$ , is the weighted mean of  $X$ 's possible values, weighted by the probability density function of those values:

$$E[X] = \int_{x \in \mathcal{R}(X)} x \cdot p(X = x) dx$$

The **expectation**, **expected value**, or **population mean** of a *discrete* random variable  $X$ , denoted  $E[X]$ ,  $\mu(X)$ , or  $\mu_X$ , is the mean of  $X$ 's possible values, weighted by the probability mass function of those values:

$$E[X] = \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x)$$

(c.f. [https://en.wikipedia.org/wiki/Expected\\_value](https://en.wikipedia.org/wiki/Expected_value))

---

**Theorem 4.3** (Expectation of the Bernoulli distribution). *The expectation of a Bernoulli random variable with parameter  $\pi$  is:*

$$E[X] = \pi$$

---

*Proof.*

$$\begin{aligned} E[X] &= \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x) \\ &= \sum_{x \in \{0,1\}} x \cdot P(X = x) \\ &= (0 \cdot P(X = 0)) + (1 \cdot P(X = 1)) \\ &= (0 \cdot (1 - \pi)) + (1 \cdot \pi) \\ &= 0 + \pi \\ &= \pi \end{aligned}$$

□

---

**Theorem 4.4** (Expectation of time-to-event variables). *If  $T$  is a non-negative random variable, then:*

$$\mu(T|\tilde{X} = \tilde{x}) = \int_{t=0}^{\infty} S(t) dt$$

---

### 4.4 Variance and related characteristics

**Definition 4.4** (Variance). The variance of a random variable  $X$  is the expectation of the squared difference between  $X$  and  $E[X]$ ; that is:

$$\text{Var}(X) \stackrel{\text{def}}{=} E[(X - E[X])^2]$$

---

**Theorem 4.5** (Simplified expression for variance).

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

---

*Proof.* By linearity of expectation, we have:

$$\begin{aligned}
\text{Var}(X) &\stackrel{\text{def}}{=} \mathbb{E}[(X - \mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2] - \mathbb{E}[2X\mathbb{E}[X]] + \mathbb{E}[(\mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\
&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2
\end{aligned}$$

□

---

**Definition 4.5** (Precision). The **precision** of a random variable  $X$ , often denoted  $\tau(X)$ ,  $\tau_X$ , or shorthand as  $\tau$ , is the inverse of that random variable's variance; that is:

$$\tau(X) \stackrel{\text{def}}{=} (\text{Var}(X))^{-1}$$

**Definition 4.6** (Standard deviation). The standard deviation of a random variable  $X$  is the square-root of the variance of  $X$ :

$$\text{SD}(X) \stackrel{\text{def}}{=} \sqrt{\text{Var}(X)}$$

---

**Definition 4.7** (Covariance). For any two one-dimensional random variables,  $X, Y$ :

$$\text{Cov}(X, Y) \stackrel{\text{def}}{=} \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

---

**Theorem 4.6.**

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$


---

*Proof.* Left to the reader.

□

---

**Lemma 4.1** (The covariance of a variable with itself is its variance). *For any random variable  $X$ :*

$$\text{Cov}(X, X) = \text{Var}(X)$$

*Proof.*

$$\begin{aligned}
\text{Cov}(X, X) &= \mathbb{E}[XX] - \mathbb{E}[X]\mathbb{E}[X] \\
&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
&= \text{Var}(X)
\end{aligned}$$

□

---

**Definition 4.8** (Variance/covariance of a  $p \times 1$  random vector). For a  $p \times 1$  dimensional random vector  $\tilde{X}$ ,

$$\begin{aligned}
\text{Var}(\tilde{X}) &\stackrel{\text{def}}{=} \text{Cov}(\tilde{X}) \\
&\stackrel{\text{def}}{=} \mathbb{E}[(\tilde{X} - \mathbb{E}\tilde{X})^\top (\tilde{X} - \mathbb{E}\tilde{X})]
\end{aligned}$$

---

**Theorem 4.7** (Alternate expression for variance of a random vector).

$$\text{Var}(X) = E[X^\top X] - E[X]^\top E[X]$$


---

*Proof.*

$$\begin{aligned} \text{Var}(X) &= E[(X^\top - E[X]^\top)(X - E[X])] \\ &= E[X^\top X - E[X]^\top X - X^\top E[X] + E[X]^\top E[X]] \\ &= E[X^\top X] - E[X]^\top E[X] - E[X]^\top E[X] + E[X]^\top E[X] \\ &= E[X^\top X] - 2E[X]^\top E[X] + E[X]^\top E[X] \\ &= E[X^\top X] - E[X]^\top E[X] \end{aligned}$$

□

---

**Theorem 4.8** (Variance of a linear combination). *For any vector of random variables  $\tilde{X} = (X_1, \dots, X_n)$  and corresponding vector of constants  $\tilde{a} = (a_1, \dots, a_n)$ , the variance of their linear combination is:*

$$\begin{aligned} \text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) \end{aligned}$$


---

*Proof.* Left to the reader...

□

---

**Corollary 4.1.** *For any two random variables  $X$  and  $Y$  and scalars  $a$  and  $b$ :*

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2(a \cdot b) \text{Cov}(X, Y)$$


---

*Proof.* Apply Theorem 4.8 with  $n = 2$ ,  $X_1 = X$ , and  $X_2 = Y$ .

Or, see <https://statproofbook.github.io/P/var-lincomb.html>

□

---

**Definition 4.9** (homoskedastic, heteroskedastic). A random variable  $Y$  is **homoskedastic** (with respect to covariates  $X$ ) if the variance of  $Y$  does not vary with  $X$ :

$$\text{Var}(Y|X = x) = \sigma^2, \forall x$$

Otherwise it is **heteroskedastic**.

---

**Definition 4.10** (Statistical independence). A set of random variables  $X_1, \dots, X_n$  are **statistically independent** if their joint probability is equal to the product of their marginal probabilities:

$$\Pr(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(X_i = x_i)$$

💡 Tip

The symbol for independence,  $\perp\!\!\!\perp$ , is essentially just  $\prod$  upside-down. So the symbol can remind you of its definition (Definition 4.10).

**Definition 4.11** (Conditional independence). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally statistically independent** given a set of covariates  $X_1, \dots, X_n$  if the joint probability of the  $Y_i$ s given the  $X_i$ s is equal to the product of their marginal probabilities:

$$\Pr(Y_1 = y_1, \dots, Y_n = y_n | X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(Y_i = y_i | X_i = x_i)$$

**Definition 4.12** (Identically distributed). A set of random variables  $X_1, \dots, X_n$  are **identically distributed** if they have the same range  $\mathcal{R}(X)$  and if their marginal distributions  $P(X_1 = x_1), \dots, P(X_n = x_n)$  are all equal to some shared distribution  $P(X = x)$ :

$$\forall i \in \{1 : n\}, \forall x \in \mathcal{R}(X) : P(X_i = x) = P(X = x)$$

**Definition 4.13** (Conditionally identically distributed). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally identically distributed** given a set of covariates  $X_1, \dots, X_n$  if  $Y_1, \dots, Y_n$  have the same range  $\mathcal{R}(X)$  and if the distributions  $P(Y_i = y_i | X_i = x_i)$  are all equal to the same distribution  $P(Y = y | X = x)$ :

$$P(Y_i = y | X_i = x) = P(Y = y | X = x)$$

**Definition 4.14** (Independent and identically distributed). A set of random variables  $X_1, \dots, X_n$  are **independent and identically distributed** (shorthand: “ $X_i$  iid”) if they are statistically independent and identically distributed.

**Definition 4.15** (Conditionally independent and identically distributed). A set of random variables  $Y_1, \dots, Y_n$  are **conditionally independent and identically distributed** (shorthand: “ $Y_i | X_i$  ciid” or just “ $Y_i | X_i$  iid”) given a set of covariates  $X_1, \dots, X_n$  if  $Y_1, \dots, Y_n$  are conditionally independent given  $X_1, \dots, X_n$  and  $Y_1, \dots, Y_n$  are identically distributed given  $X_1, \dots, X_n$ .

## 4.5 The Central Limit Theorem

The sum of many independent or nearly-independent random variables with small variances (relative to the number of RVs being summed) produces bell-shaped distributions.

For example, consider the sum of five dice (Figure 4).

```
library(dplyr)
dist =
  expand.grid(1:6, 1:6, 1:6, 1:6, 1:6) |>
  rowwise() |>
  mutate(total = sum(c_across(everything())))) |>
  ungroup() |>
  count(total) |>
  mutate(`p(X=x)` = n/sum(n))

library(ggplot2)
```

```
dist |>
  ggplot() +
  aes(x = total, y = `p(X=x)`) +
  geom_col() +
  xlab("sum of dice (x)") +
  ylab("Probability of outcome, Pr(X=x)") +
  expand_limits(y = 0)
```

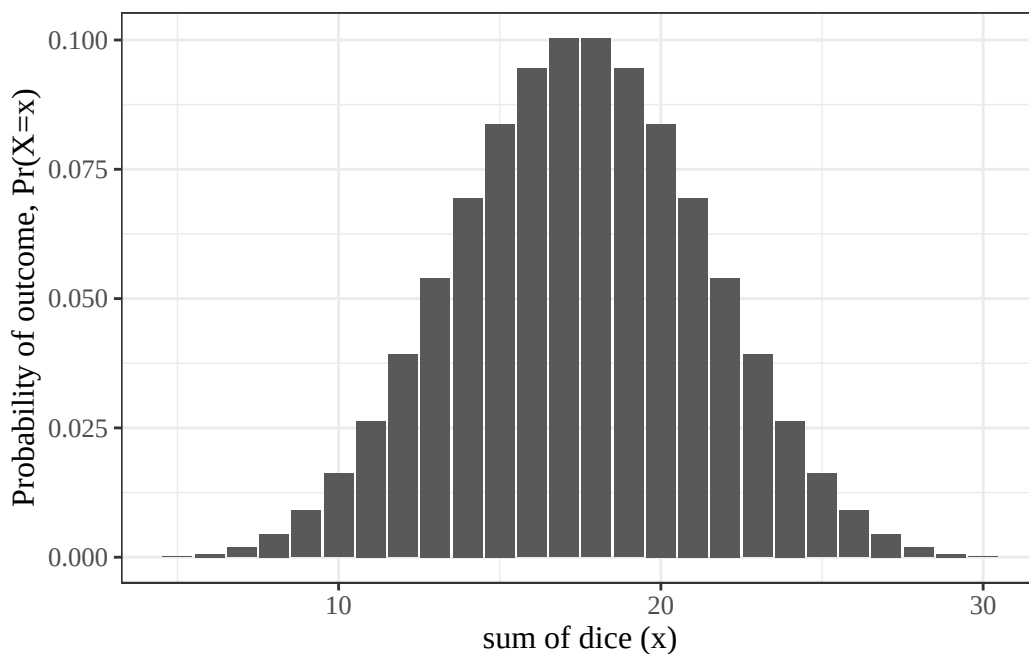


Figure 4: Distribution of the sum of five dice

In comparison, the outcome of just one die is not bell-shaped (Figure 5).

```
library(dplyr)
dist =
  expand.grid(1:6) |>
  rowwise() |>
  mutate(total = sum(c_across(everything())) |>
  ungroup() |>
  count(total) |>
  mutate(`p(X=x)` = n/sum(n))

library(ggplot2)

dist |>
  ggplot() +
  aes(x = total, y = `p(X=x)`) +
  geom_col() +
  xlab("sum of dice (x)") +
  ylab("Probability of outcome, Pr(X=x)") +
  expand_limits(y = 0)
```

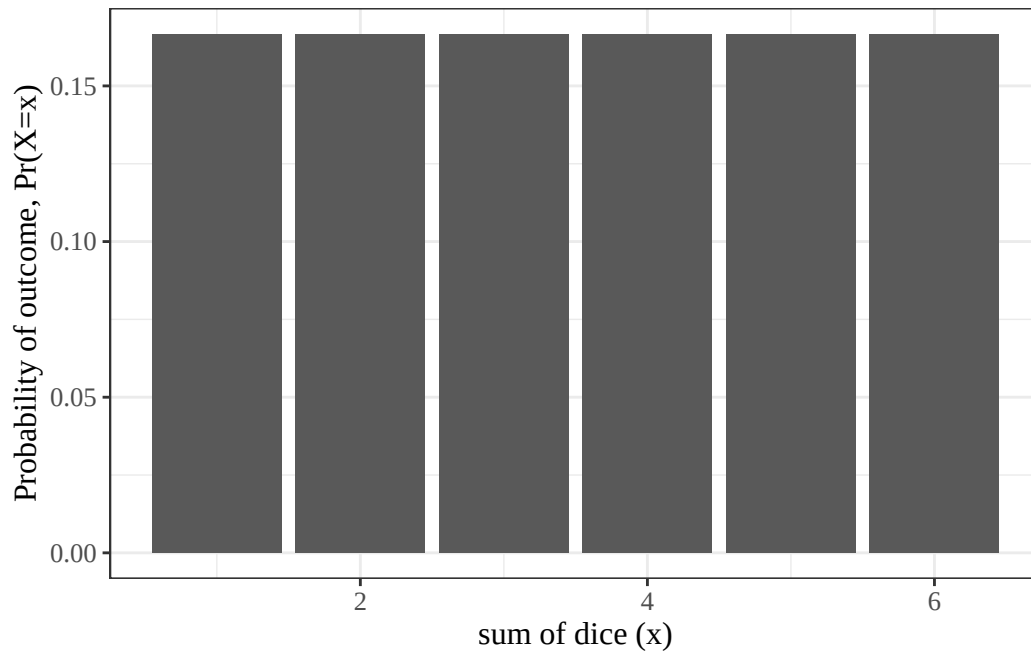


Figure 5: Distribution of the outcome of one die

What distribution does a single die have?

Answer: discrete uniform on 1:6.

## 5 Additional resources

- Miller (2017)

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.

Kalbfleisch, John D, and Ross L Prentice. 2011. *The Statistical Analysis of Failure Time Data*. John Wiley & Sons.

Klein, John P, and Melvin L Moeschberger. 2003. *Survival Analysis: Techniques for Censored and Truncated Data*. Vol. 1230. Springer. <https://link.springer.com/book/10.1007/b97377>.

Kleinbaum, David G, and Mitchel Klein. 2012. *Survival Analysis: A Self-Learning Text*. 3rd ed. Springer. <https://link.springer.com/book/10.1007/978-1-4419-6646-9>.

Miller, Steven J. 2017. *The Probability Lifesaver : All the Tools You Need to Understand Chance*. A Princeton Lifesaver Study Guide. Princeton University Press. <https://press.princeton.edu/books/hardcover/9780691149547/the-probability-lifesaver>.

Rothman, Kenneth J., Timothy L. Lash, Tyler J. VanderWeele, and Sebastien Haneuse. 2021. *Modern Epidemiology*. Fourth edition. Wolters Kluwer.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.